

Docker Assignment-2

Docker Volumes:

```
python-backend > Dockerfile
1 FROM python:3.11-slim

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend> cd python-backend
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker volume create myvolume
myvolume
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker volume ls
DRIVER      VOLUME NAME
local       myvolume
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker run -v myvolume:/data -d -p python-backend:v1

What's next:
  Debug this container error with Gordon → docker ai "help me fix this container error"
docker: 'docker run' requires at least 1 argument

Usage: docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

See 'docker run --help' for more information
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker run -v myvolume:/data -d -p 80:8000 python-backend:v1
9218b79bb441ecca1d98e0cc99e18e2c947d17af8fcd8598d5dc09d86e1316e

What's next:
  Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint
t stupefied_aryabhata (c0b7350f30fc97459dbde51051b1580d91ba072605fa60854c91a393847c47ed): Bind for 0.0.0.0:80 failed: port is already
allocated

Run 'docker run --help' for more information
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
NAMES					

```
Run 'docker run --help' for more information
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS        PORTS
NAMES
6166e12a8f7e   python-backend:v1 "uvicorn main:app --..." 7 minutes ago  Up 7 minutes  0.0.0.0:80->8000/tcp, [::]:80->8000/tcp
magical_benz
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker rm -f 6166e12a8f7e
6166e12a8f7e
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS        PORTS
NAMES
0249a113cf7a   python-backend:v1 "uvicorn main:app --..." 7 seconds ago  Up 6 seconds  0.0.0.0:80->8000/tcp, [::]:80->8000/tcp
nice_sutherland
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker exec -it 0249a113cf7a sh
# ls
Dockerfile  __pycache__  main.py      models.py    requirements.txt
# cd /data
# ls
# cd /app
# ls
Dockerfile  __pycache__  main.py      models.py    requirements.txt
# exit
```

```
EXPLORER
PYTHON-BACKEND
python-backend
  __pycache__
  Dockerfile
  main.py
  models.py
  requirements.txt

python-backend > Dockerfile
1 FROM python:3.11-slim

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
docker + - - - - -

Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker rm -f 0249a113cf7a
0249a113cf7a
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker run -v myvolume:/app/data -d -p 80:8000 python-backend:v1
252f86133a94ee0a5070b171761df6211fc20e074ae3b66a2866d5858f22f80c
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS        PORTS
NAMES
252f86133a94   python-backend:v1 "uvicorn main:app --..." 8 seconds ago  Up 7 seconds  0.0.0.0:80->8000/tcp, [::]:80->8000/tcp
inspiring_taussig
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker exec -it 252f86133a94 sh
# ls
Dockerfile __pycache__ data main.py models.py requirements.txt
# cd data
# echo "HELLO DOCKER VOLUME">SAMPLE1.txt
# ls
SAMPLE1.txt
# cat SAMPLE1.txt
HELLO DOCKER VOLUME
# exit

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug 252f86133a94
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker rm -f 252f86133a94
252f86133a94
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker run -v myvolume:/app/data -d -p 80:8000 python-backend:v1
3a4105dda2f9443b90a8fd467830331b92621f788a11e3c961b2d057f4b54416
```

```
252f86133a94
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker run -v myvolume:/app/data -d -p 80:8000 python-backend:v1
3a4105dda2f9443b90a8fd467830331b92621f788a11e3c961b2d057f4b54416
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS        PORTS
NAMES
3a4105dda2f9   python-backend:v1 "uvicorn main:app --..." 9 seconds ago  Up 8 seconds  0.0.0.0:80->8000/tcp, [::]:80->8000/tcp
sharp_yalow
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker exec -it 3a4105dda2f9 sh
# ls
Dockerfile __pycache__ data main.py models.py requirements.txt
# cd data
# ls
SAMPLE1.txt
#
```

```
python-backend 1 FROM python:3.11-slim
> __pycache__
saro
Bind-Mount
Dockerfile
main.py
models.py
requirements.txt

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> mkdir saro

Directory: C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend

Mode                LastWriteTime         Length Name
----                -
d-----          27-07-2026      19:28             saro

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> cd saro
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend\saro> pwd

Path
----
C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend\saro

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend\saro> docker run -v "C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend\saro:/app/data" -d -p 80:8000 python-backend:v1
8e6327237169bd737ad757e079da802ff4dc615131bfac7ca8727a8bdadc3c
```

```
3bb60bde562e python-backend:v1 "uvicorn main:app --..." About a minute ago Up About a minute 0.0.0.0:80->8000/tcp, [::]:80->8000/tcp tender_mcclintock
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
NAMES
3bb60bde562e python-backend:v1 "uvicorn main:app --..." About a minute ago Up About a minute 0.0.0.0:80->8000/tcp, [::]:80->8000/tcp tender_mcclintock
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend\saro> docker exec -it 3bb60bde562e sh
# ls
Dockerfile __pycache__ data main.py models.py requirements.txt
# cd data
# ls
Bind-Mount
# cat Bind-Mount
Hello World#
```